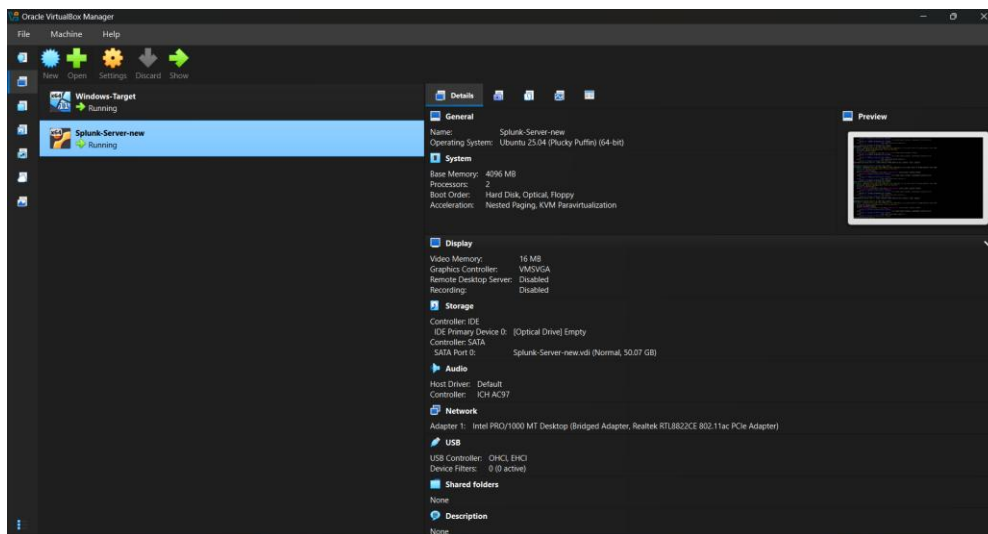


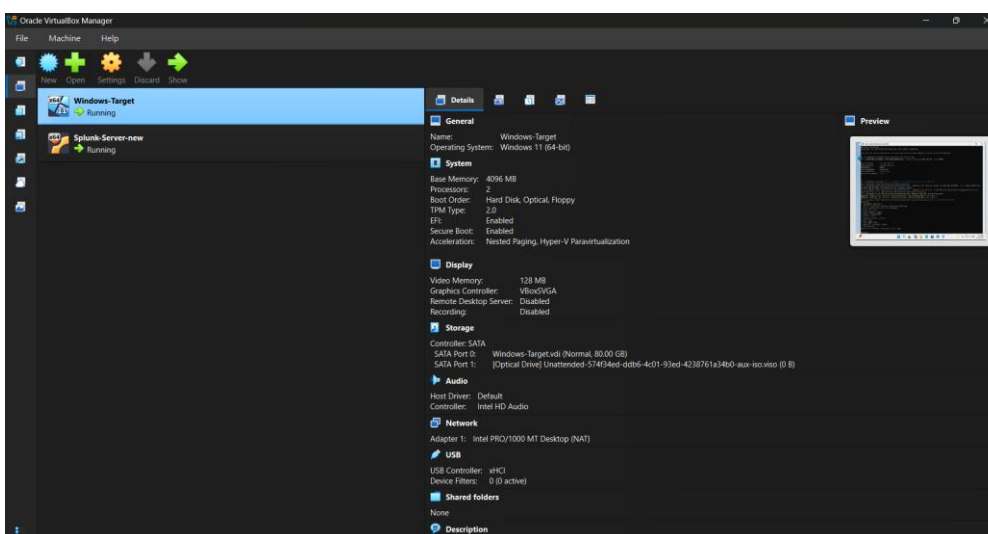
SPLUNK HOME LAB: CROSS-PLATFORM DATA INGESTION

Environment:

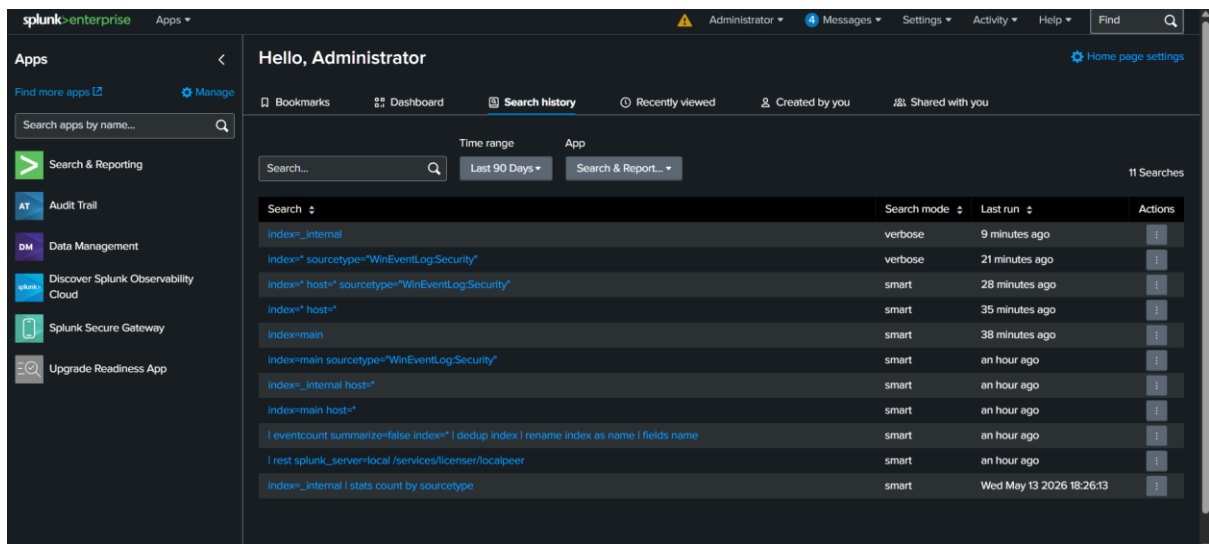
- Indexer OS: Ubuntu 22.04 LTS
- Forwarder OS: Windows 11 Pro
- Total Events Indexed: 121,678+
- Time zone: Asia/ Kuala Lumpur



Ubuntu 22.04 LTS



Windows 11 Pro



SPLUNK

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> Start-Service SplunkForwarder
PS C:\WINDOWS\system32> Test-NetConnection -ComputerName 10.248.186.89 -Port 9997

ComputerName      : 10.248.186.89
RemoteAddress     : 10.248.186.89
RemotePort        : 9997
InterfaceAlias    : Ethernet
SourceAddress     : 10.0.2.15
TcpTestSucceeded  : True
```

Figure 1: Verification of Cross-Platform Network Connectivity

Using the Test-Net Connection cmdlet in Windows PowerShell, I validated that port 9997 on the Ubuntu Indexer (10.248.186.89) is reachable. The TcpTestSucceeded: True result confirms that the virtual network routing and the Ubuntu UFW firewall allow for seamless data transmission.

```
GNU nano 8.7.1
[general]
allowRemoteLogin = always
[general]
serverName = Splunk-Server-new
pass4SymmKey = $7$XJbjuRJwyxbykHznK7Z2msoounZubUV2P4HbLXxti9Tnb3Rq9xnfg==

[sslConfig]
sslPassword = $7$d5/CrJW0ho9o/hhpdYIRXi0W55tzezqPwxNtw3FPvgzGNlMyEe9+og==

[lmppool:auto_generated_pool_download-trial]
description = auto_generated_pool_download-trial
peers = *
quota = MAX
stack_id = download-trial

[lmppool:auto_generated_pool_forwarder]
description = auto_generated_pool_forwarder
peers = *
quota = MAX
stack_id = forwarder

[lmppool:auto_generated_pool_free]
description = auto_generated_pool_free
peers = *
quota = MAX
stack_id = free
```

Figure 2: Indexer Access Configuration

Modified the server.conf file on the Ubuntu Indexer using the Nano text editor. By setting allowRemoteLogin to always, I enabled the administrative capability for the remote Windows Universal Forwarder to authenticate and establish a persistent data stream.

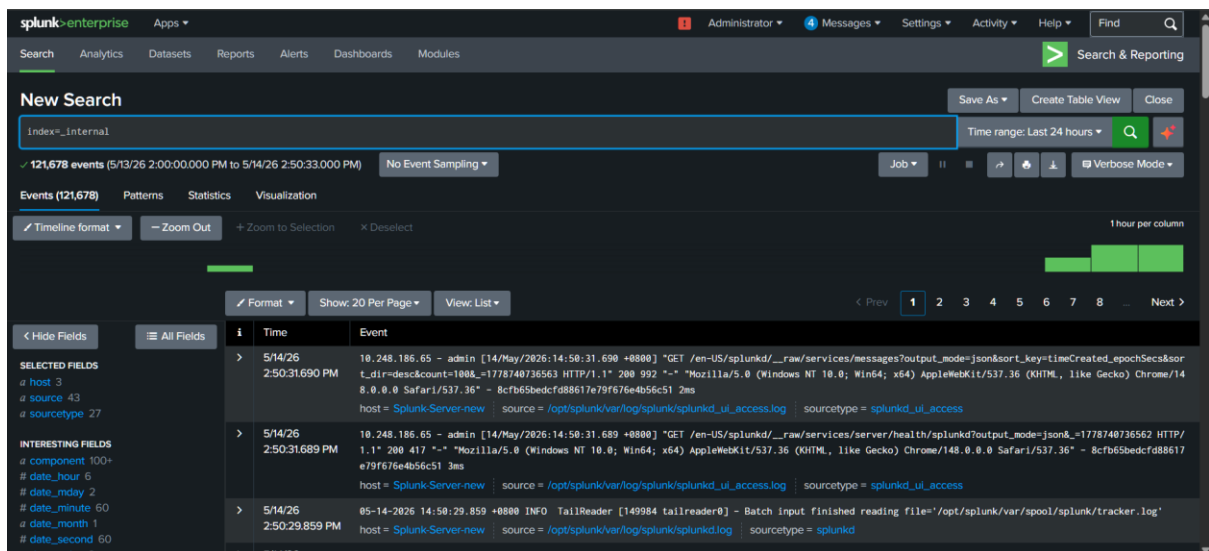


Figure 3: Production-Level Ingestion and Temporal Accuracy

Final validation of the SIEM pipeline confirms 121,678 active events successfully indexed. The real-time data stream shows perfect synchronization with the local system clock (2:50 PM), validating that both the log transmission and the time-zone configuration (Asia/Kuala Lumpur) are fully optimized for security operations.

1. Issues Faced & Resolutions

- **Issue: Authentication Failure (401 Unauthorized)**
- Resolution: Manually modified server.conf on the Ubuntu Indexer to set allowRemoteLogin = always, enabling the Windows VM to "talk" to the Linux backend.
- **Issue: Data Visibility (0 Events Found)**
- Resolution: Identified an 8-hour time synchronization offset between the UTC server and the local machine. Fixed by standardizing the system time zone to Asia/Kuala Lumpur.
- **Issue: Network Handshake Failure**
- Resolution: Configured the Ubuntu UFW (Firewall) to permit ingress traffic on Port 9997 and verified the connection using the Test-NetConnection cmdlet.

2. Technical Environment (The "Specs")

- **Virtualization:** Oracle VM VirtualBox
- **Indexer OS:** Ubuntu 22.04 LTS (Linux)
- **Endpoint OS:** Windows 11 Pro (Virtual Machine)
- **Software:** Splunk Enterprise (Server) & Universal Forwarder (Agent)
- **Communication Port:** TCP 9997

3. Skills Demonstrated

- **Linux CLI:** Skilled in bash navigation and using nano for configuration management.
- **Windows SysAdmin:** Experienced in PowerShell administration and managing Windows Services.
- **SIEM Ingestion:** Ability to configure Universal Forwarders and define data inputs/indexes.
- **Network Troubleshooting:** Proficient in diagnosing connectivity issues using netstat, ping, and port scanning.

SUMMARY

This SPLUNK HOME LAB project successfully demonstrated the deployment of a cross-platform log management system. By troubleshooting network latency, firewall restrictions, and time-sync offsets, a robust connection was established between a Windows 11 endpoint and a centralized Ubuntu Indexer. The environment is now prepared for advanced threat hunting and security monitoring.